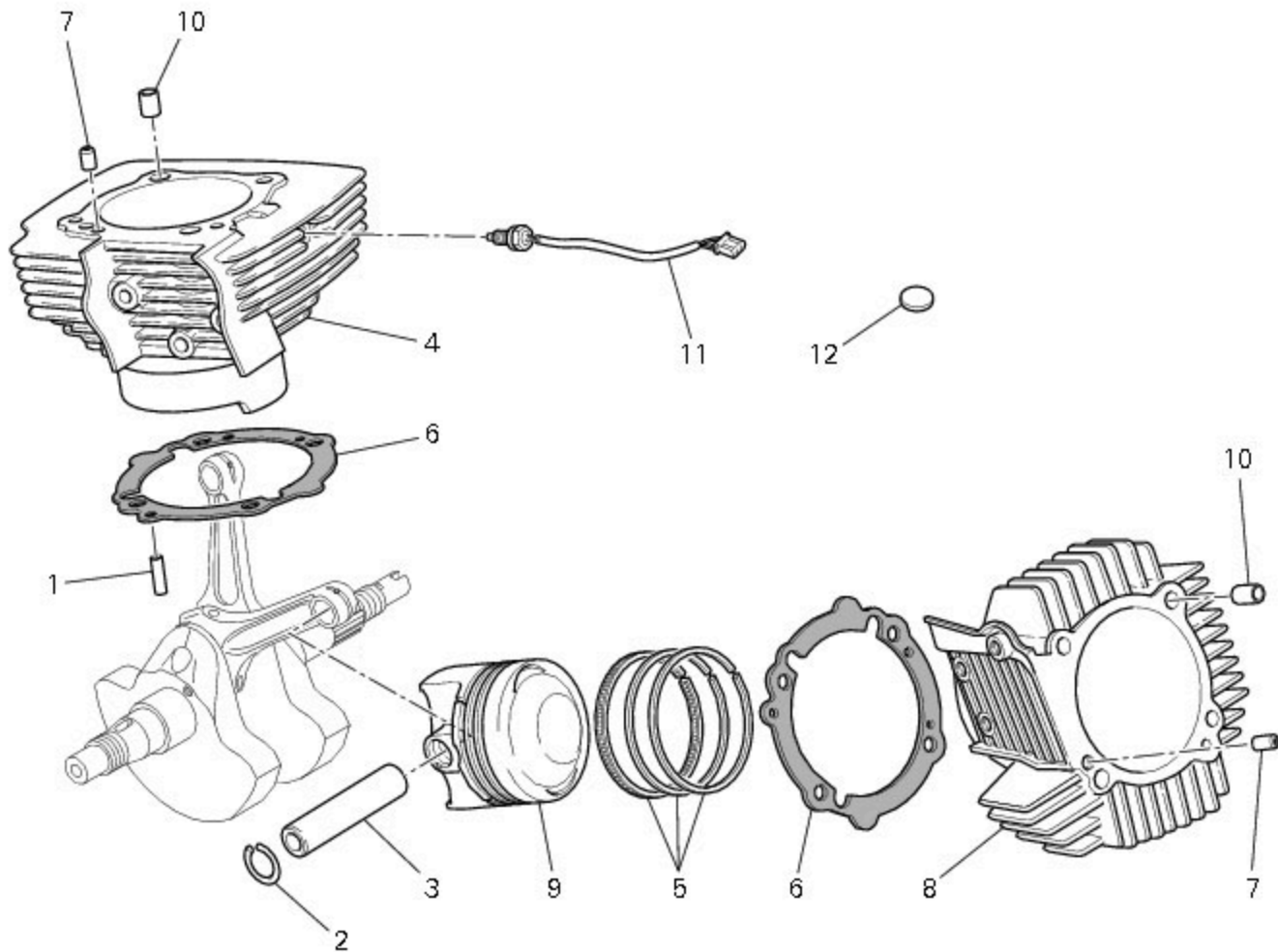


5 -Cylinder/piston assemblies



- 1 Pin
- 2 Circlip
- 3 Gudgeon pin
- 4 Vertical cylinder
- 5 Piston rings
- 6 Cylinder-crankcase gasket
- 7 Pin
- 8 Horizontal cylinder
- 9 Piston
- 10 Centring bush
- 11 Temperature sensor
- 12 Vibration damper mount

 Spare parts catalogue

1100 ABS [CYLINDERS - PISTONS](#)
 1100 S ABS [CYLINDERS - PISTONS](#)

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text. Refer to the exploded view diagram.

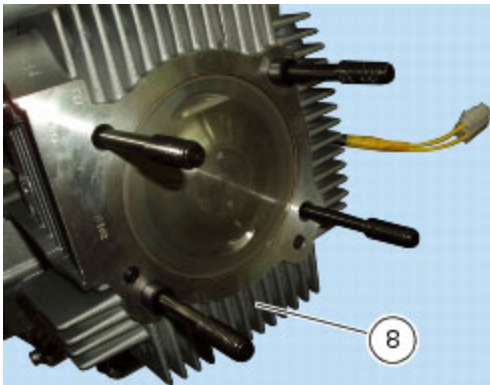
Removal of the cylinder/piston assembly

Operation	Section reference
Remove the engine from the frame	N 1, Removal of the engine
Remove the side timing belt covers	N 4.2, Removal of the timing belt covers
Remove the timing belts	N 4.2, Removal of the timing system assembly
Remove the cylinder heads from the engine	N 4.4, Removal of the cylinder head assembly

Remove the dowel (7), bushes (10) and head gasket (A) (Sect. N 4.4, [Removal of the cylinder head assembly](#)) from their seats on the cylinder barrel.

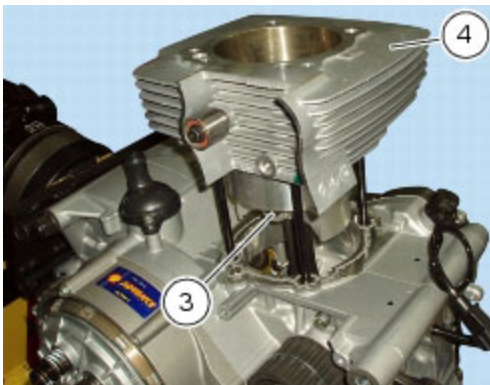


Use the tool no. **88713.1805** to bring the horizontal cylinder piston (8) close to TDC.



Carefully lift the cylinder barrel (4) off the crankcase, keeping it vertical.

If necessary, rock the cylinder slightly using both hands or tap its base gently with a rubber mallet. Continue to lift the cylinder until you can access the gudgeon pin (3).



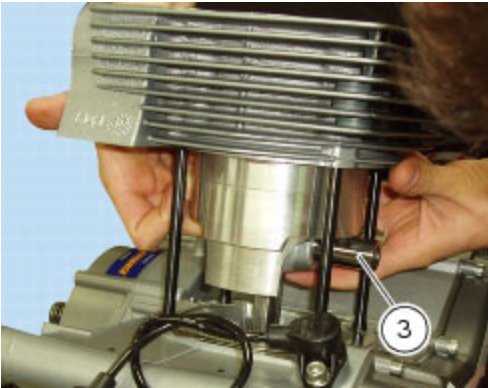
Since insertion of piston in the barrel is a difficult operation to perform at the time of reassembly, remove the piston together with the barrel as an assembly as described below.

Close the opening in the engine crankcase with a rag or soft paper to prevent foreign material or solid combustion residue from falling in.

Remove the circlip (2) of the gudgeon pin (3) from the clutch side.



Working from the opposite side, drive out the gudgeon pin (3) sufficiently to release the connecting rod.



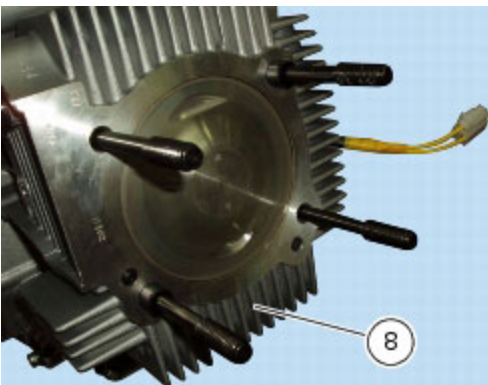
Lift the cylinder/piston assembly clear of the engine crankcase studs.
If work is to be carried out on the piston, carefully withdraw it from the cylinder.
Remove cylinder-crankcase gaskets (6) from the cylinder barrel.

Important

Mark the pistons to show from which cylinder they were removed:
V = Vertical - H = Horizontal.



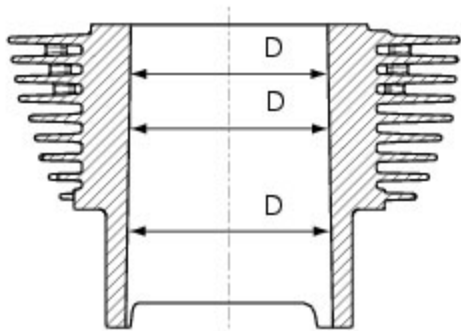
Use service tool no. **88713.1805** to bring the horizontal cylinder piston (8) close to TDC. To remove the vertical cylinder-piston assembly, proceed as described above for the horizontal cylinder.



Overhaul of the cylinder barrel/piston components

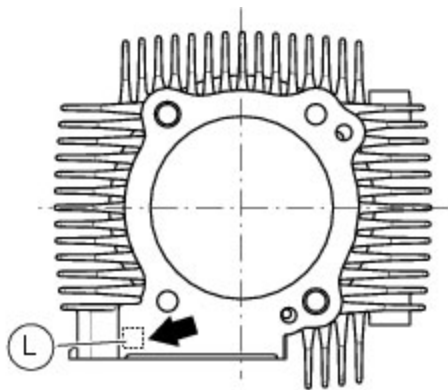
Overhauling the cylinder

Check that the walls of the cylinder bore are perfectly smooth. Measure the cylinder bore diameter (D) at three different heights and in two directions offset by 90°, thereby determining the assembly clearance, taper, and ovality (see specified values in Section C 1.1, [Cylinder/Piston](#)).

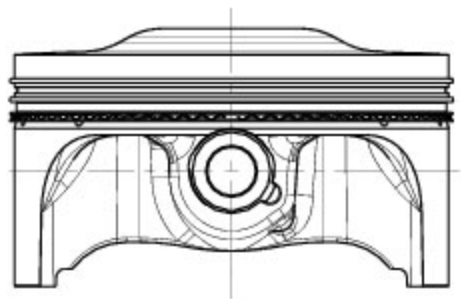


In the event of damage or excessive wear the barrel must be renewed as it has a silicon carbide coating (which provides the cylinder walls with excellent anti-friction and anti-wear properties) and therefore cannot be rebored. The cylinders are marked by a letter (L) (punched into the specific surface on the cylinder side) showing the size class to which they belong (A, B or C).

Always match cylinders with pistons from the same size class.



Overhaul of the piston

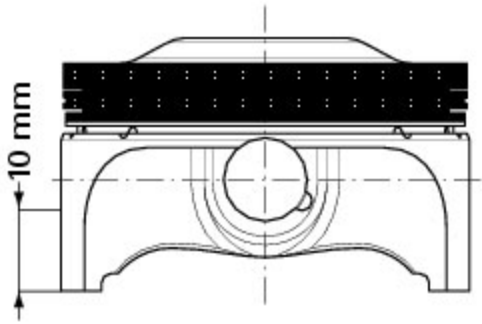


Clean the piston crown and piston ring grooves, removing any carbon deposits.

Inspect the piston and check its dimensions carefully: there must be no signs of scuffing, scoring, cracks, or other damage.

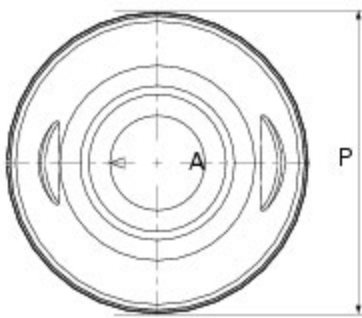
The piston diameter (P) must be measured at **10** mm from the skirt base at right angles to the axis of the gudgeon pin.

The pistons must always be renewed as a pair.



Piston-cylinder clearance

The pistons are marked with a letter (punched into the piston crown) that indicates the size class to which they belong. Always match cylinders with pistons from the same size class. The prescribed values are given in Sect. C 1.1, [Cylinder/Piston](#).



Overhauling the gudgeon pins



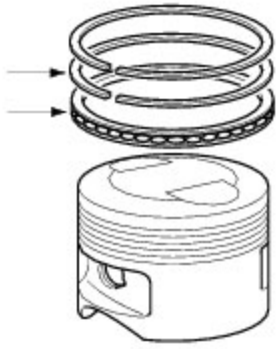
Gudgeon pins must be perfectly smooth without signs of scoring, steps, or blueing due to overheating. The well-lubricated gudgeon pin must slide smoothly inside the piston without stiffness.

The assembly clearance with the piston at the time of installation must be **0.002** to **0.008** mm. the maximum permissible wear limit is **0.025** mm.

If a new gudgeon pin is fitted, you must also change the connecting rod small end bush.

Overhauling the piston rings

The piston rings must not show any signs of scuffing or scoring. Replacement pistons are supplied complete with piston rings and gudgeon pin.



Piston ring-to-groove clearance

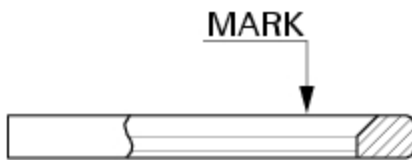
The maximum permissible wear limit is **0.15** mm for the top segment (1st) and **0.10** mm for the others (2nd and oil scraper ring).



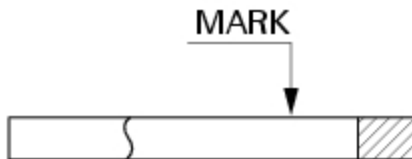
Note

The piston rings must always be installed with markings facing upwards.

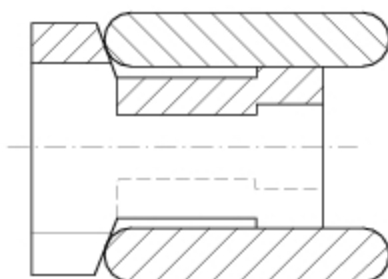
1st segment



2nd segment



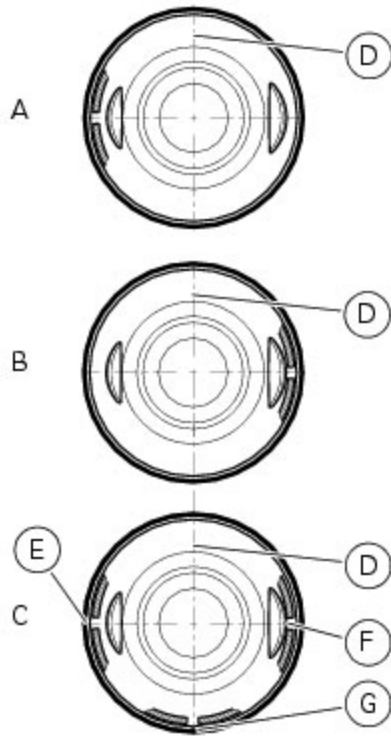
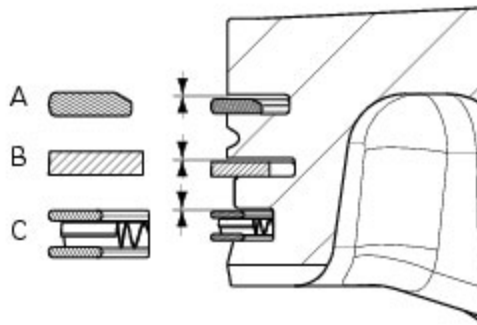
oil scraper ring



The positioning of the piston ring opening with respect to valve seats is shown below:

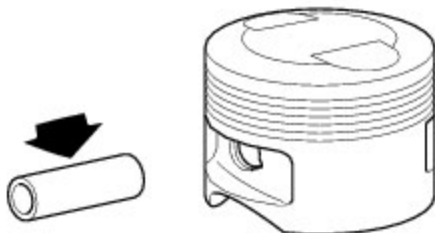
A 1st groove

- B 2nd groove
- C 3rd groove
- D piston axis
- E top ring opening
- F bottom ring opening
- G spring ring opening

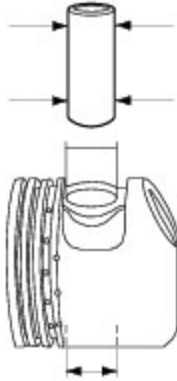


Gudgeon pin-piston clearance

Measure the diameter of the gudgeon pin bore in the piston.



Measure the diameter of the gudgeon pin.

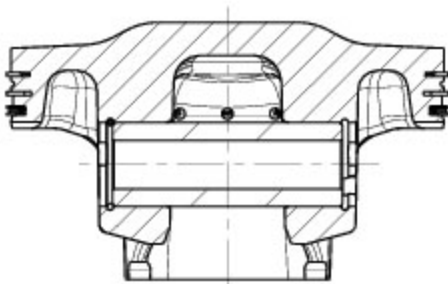


Maximum wear limit:
0.025 mm.



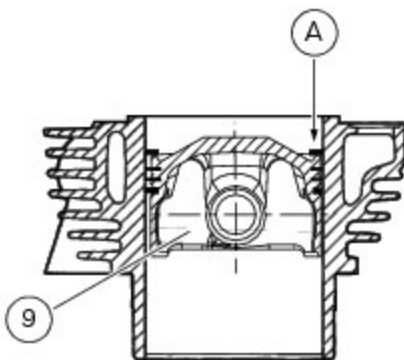
Note

If a new gudgeon pin is fitted, you must also change the connecting rod small end bush.



Piston ring-to-bore clearance

It is essential that the measurement be performed with the piston ring (A) positioned perfectly square in the cylinder. For this reason, the piston (9) should be inserted in the cylinder without piston rings fitted and then place the piston ring to be checked on top of the piston.



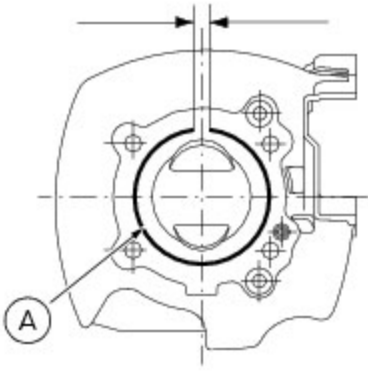
Measure the piston ring gap.

Top and second piston ring (nominal): **0.20 to 0.40 mm.**

Wear limit: **0.80 mm.**

Oil control ring (nominal): **0.30 to 0.60 mm.**

Wear limit: **1.0 mm.**

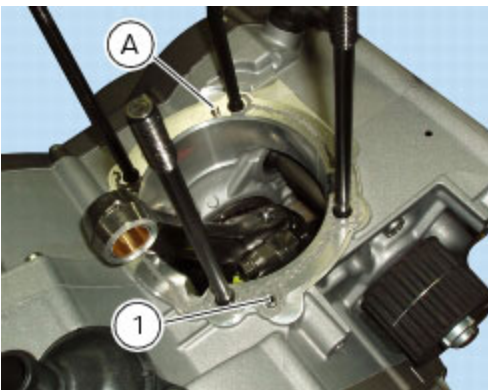


Refitting the cylinder/piston assembly

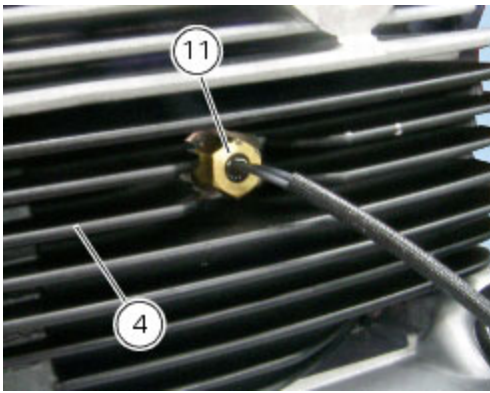
If a new cylinder/piston assembly is to be fitted, make sure that the piston and cylinder are of the same size class. The letter indicating the cylinder size class is stamped on the cylinder-head mating face; the piston size class letter is stamped on the crown in between the two recesses.



If the pistons have been separated from their cylinders, before reassembling these components, position the piston ring gaps at 120° from one another (the markings must always face the piston crown). Use a universal tool to carefully insert the piston into the cylinder (first lubricate the inside of the cylinder with engine oil). Position the cylinder with the smallest valve pockets facing the exhaust port. Remove any deposits and degrease the contact surfaces of the crankcase half and the cylinders. Check that connection (1) for transfer of oil to the cylinder heads and cylinder locating dowel (A) are installed on the crankcase.



The vertical cylinder temperature sensor (11) is located on the vertical cylinder (4).



Apply a bead of liquid gasket to the mating surfaces of gasket (6):
the exact configuration of the bead of liquid gasket varies according to whether the gasket is for the (6) for the vertical cylinder (V) and the horizontal cylinder (O). The picture shows liquid gasket application surfaces for the various configurations.

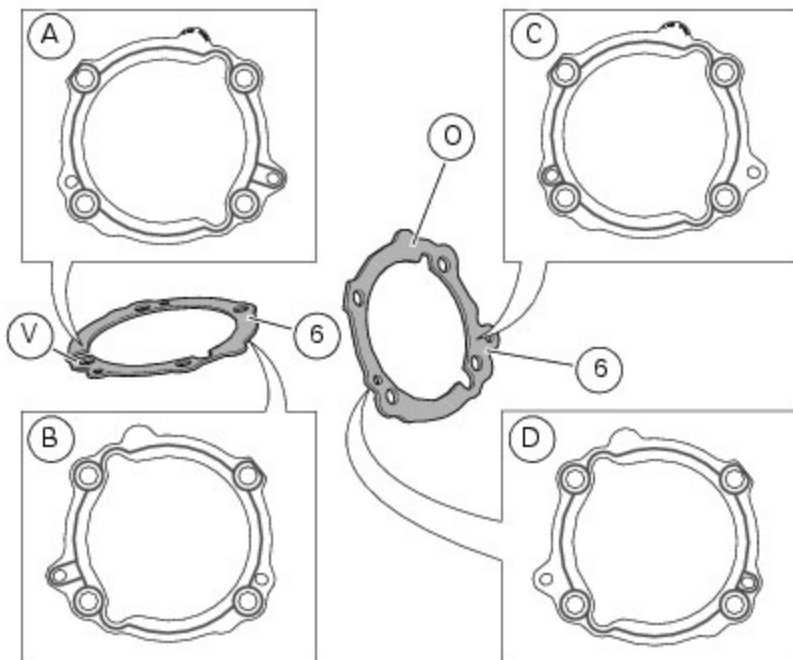
Surface in contact with the vertical cylinder (A).

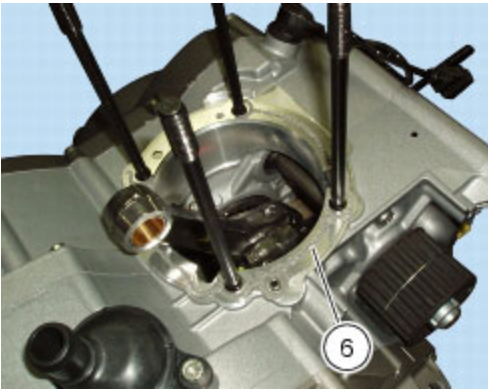
Surface in contact with the crankcase, at the vertical cylinder (B).

Surface in contact with the horizontal cylinder (C).

Surface in contact with the crankcase, at the horizontal cylinder (D).

Locate the gasket (6) on the crankcase with the word "TOP" facing upwards.

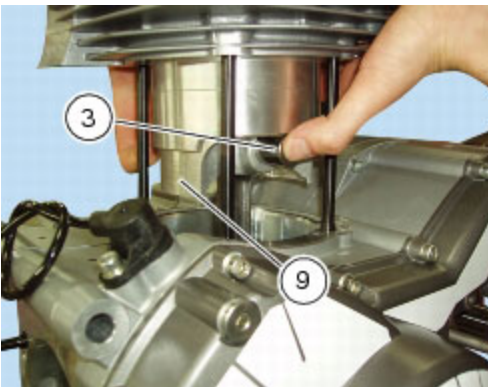




Using service tool no. **88713.1805**, bring the small end of the connecting rod small end up to near TDC. Fit the piston-cylinder assembly on the crankcase studs.



Push the connecting rod small end into the piston close to the gudgeon pin (3) bore. Lubricate and insert the gudgeon pin. The gudgeon pin must slide smoothly in the connecting rod small end bush and in the piston (9).



Stuff the crankcase opening with a cloth to prevent foreign objects from falling inside and then fit the circlip (2).

Important

Always fit new circlips (2) on reassembly.



Push the cylinder barrel (4) down until it seats against the crankcase.



Important

Fit the pilot pin (7), head gasket (A) (Sect. N 4.4, [Refitting the cylinder heads](#)) and locating dowel (10).



Repeat the procedure for the other cylinder.

Operation	Section reference
Refit the cylinder heads	N 4.4, Refitting the cylinder head assembly
Refit the timing belts	N 4.2, Refitting the timing belt covers
Refit the timing belt covers	N 4.2, Refitting the timing belt covers
Refit the engine to the frame	N 1, Refitting the engine